

CMP-06 - Attention Manager

Admit, prioritize, place, and preempt bounded work under declared resource, urgency, fairness, and constitutional constraints.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 08. Authoritative state: Attention Objects, work-admission decisions, priority and preemption records, placement leases. Explicitly non-owned: Intent, Judgment, resource inventory, action permission.

Critical invariant

The Attention Manager is the sole authority for work admission and placement; urgency cannot bypass constitutional or authority controls.

Logical interfaces

CMP-06-IF-01 / Command / Submit Attention Request

Command `Submit Attention Request` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-02 / Command / Admit Work

Command `Admit Work` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-03 / Command / Preempt Work

Command `Preempt Work` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-04 / Command / Release Placement

Command `Release Placement` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-05 / Query / Get Queue State

Query `Get Queue State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-06 / Query / Explain Priority

Query `Explain Priority` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-07 / Query / Get Placement

Query `Get Placement` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-08 / Event / Work Admitted

Event `Work Admitted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-09 / Event / Work Preempted

Event `Work Preempted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-06-IF-10 / Event / Placement Released

Event `Placement Released` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Requested > Admitted > Running > Preempted, Completed, or Rejected. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: priority inversion, starvation, resource overcommit, stale lease, unbounded queue. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-06-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-06-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-06-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-06-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-06-TST-005 - Critical invariant. The Attention Manager is the sole authority for work admission and placement; urgency cannot bypass constitutional or authority controls. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-06-TST-006 - Failure containment. Inject priority inversion, starvation, resource overcommit, stale lease, unbounded queue. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-06-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-06-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-06-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-06-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.